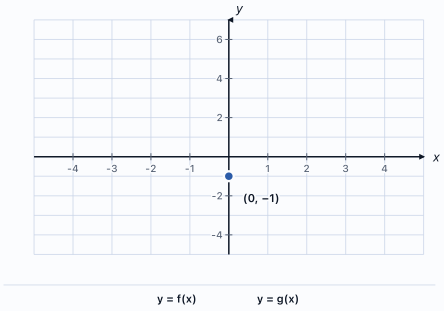
25 Structured Exam-Style Questions **Name:** _____ **Date:** _____ **Score:** _____ / 25

Differentiation, Stationary Points & Kinematics Practice Exercises

1 MCQ

The coordinate diagram shows the parabolic curve $f_1(x) = x^2 - 18x + 98$ in the domain of Differentiation, Stationary Points & Kinematics. By completing the square $f_1(x) = (x-h)^2 + k$, find the sum $k + f_1(11)$.



A 47
B 21
C 38
D 36

2 MCQ

A mathematical model in Differentiation, Stationary Points & Kinematics is defined by the cubic function $y_2(x) = 2x^3 - 9x^2 + 15x - 14$. Calculate the exact gradient of the tangent to the curve at the point where $x = 3$.

A 69
B 0
C 15 + 1
D 15

3 MCQ

In Differentiation, Stationary Points & Kinematics, two displacement vectors are given by $\mathbf{u}_3 = \langle 9, -8 \rangle$ and $\mathbf{v}_3 = \langle 9, 7 \rangle$. Compute the exact scalar value of $\mathbf{u}_3 \cdot \mathbf{v}_3 + \|\mathbf{u}_3\|^2$.

A 170
B 25
C 145
D 282

4 MCQ

The linear growth profile shown in the diagram for Differentiation, Stationary Points & Kinematics follows an arithmetic progression with first term $a_1 = 10$ and common difference $d = 6$. Find the exact sum S_{15} of the first 15 terms.

A 94

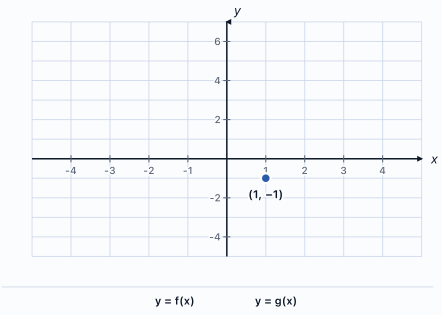
B 780

C 870

D 1410

5 MCQ

In Differentiation, Stationary Points & Kinematics, evaluate the exact definite integral $I_5 = \int_0^2 (18x^2 + 36x)\,\mathrm{d}x$.



A 144

B 156

C 120

D 61

6 MCQ

In Differentiation, Stationary Points & Kinematics, consider the quadratic function $f_6(x) = x^2 - 20x + 189$. Express $f_6(x)$ in the completed-square form $(x - h)^2 + k$ and compute the exact value of $k + f_6(17)$.

A 237

B 138

C 220

D 227

7 MCQ

The graph illustrates the cubic model $y_7(x) = 2x^3 - 10x^2 + 33x - 21$ studied in Differentiation, Stationary Points & Kinematics. Find the instantaneous gradient $\frac{\mathrm{d}y}{\mathrm{d}x}$ of the tangent to the curve at $x = 8$.

A 257

B 417

C 224

D 61

8 MCQ

In Differentiation, Stationary Points & Kinematics, two displacement vectors are given by $\mathbf{u}_8 = \langle 10, -7 \rangle$ and $\mathbf{v}_8 = \langle 8, 5 \rangle$. Compute the exact scalar value of $\mathbf{u}_8 \cdot \mathbf{v}_8 + \|\mathbf{u}_8\|^2$.

A 45

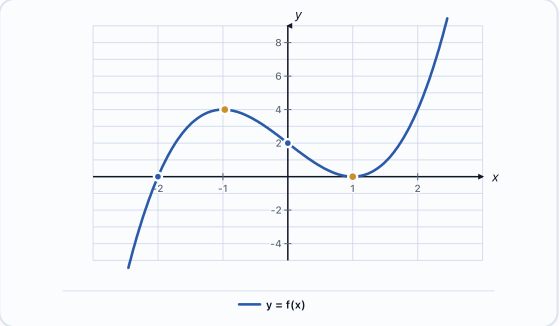
B 194

C 149

D 264

9 MCQ

Find the coordinates of the local minimum stationary point on the curve $y = x^3 - 3x + 2$ for $x > 0$ (Curve #1).



A $(1, 2)$

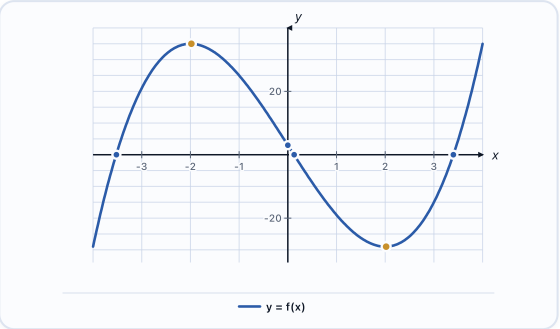
B $(2, -1)$

C $(1, 0)$

D $(-1, 0)$

10 MCQ

Find the coordinates of the local minimum stationary point on the curve $y = 2x^3 - 24x + 3$ for $x > 0$ (Curve #2).



A $(-2, -29)$

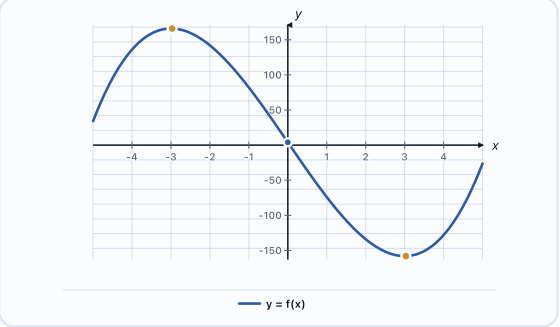
B $(2, 3)$

C $(2, -29)$

D $(4, -31)$

11 MCQ

Find the coordinates of the local minimum stationary point on the curve $y = 3x^3 - 81x + 4$ for $x > 0$ (Curve #3).



A $(6, -161)$

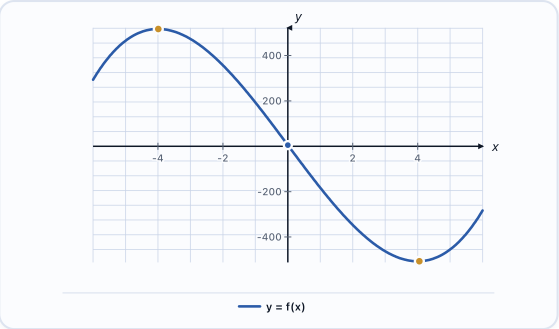
B $(-3, -158)$

C $(3, -158)$

D $(3, 4)$

12 MCQ

Find the coordinates of the local minimum stationary point on the curve $y = 4x^3 - 192x + 5$ for $x > 0$ (Curve #4).



A $(4, -507)$

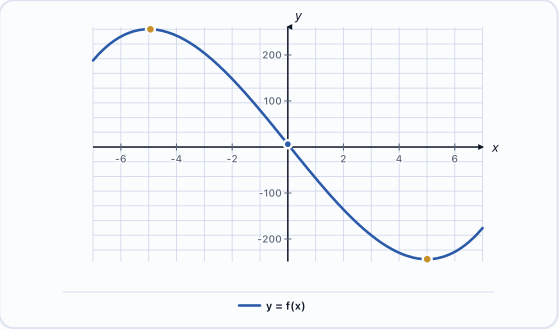
B $(8, -511)$

C $(-4, -507)$

D $(4, 5)$

13 MCQ

Find the coordinates of the local minimum stationary point on the curve $y = x^3 - 75x + 6$ for $x > 0$ (Curve #5).



A $(-5, -244)$

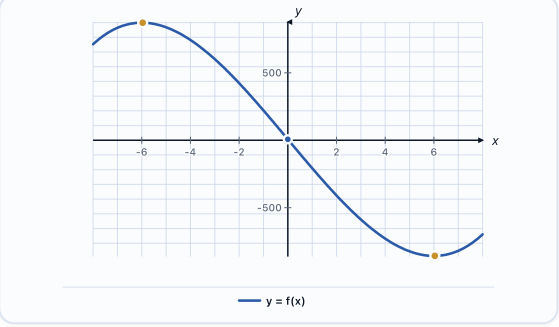
B $(5, 6)$

C $(5, -244)$

D $(10, -245)$

14 MCQ

Find the coordinates of the local minimum stationary point on the curve $y = 2x^3 - 216x + 7$ for $x > 0$ (Curve #6).



A $(6, 7)$

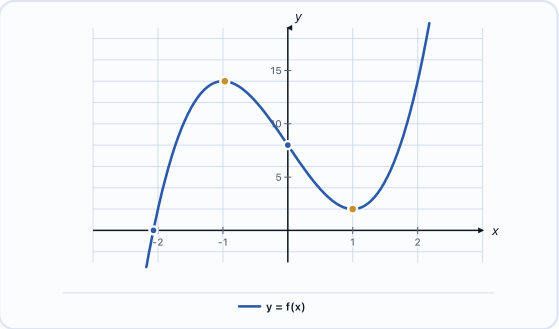
B $(6, -857)$

C $(-6, -857)$

D $(12, -859)$

15 MCQ

Find the coordinates of the local minimum stationary point on the curve $y = 3x^3 - 9x + 8$ for $x > 0$ (Curve #7).



A $(1, 2)$

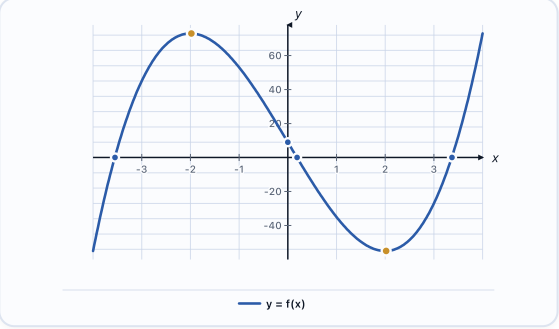
B $(1, 8)$

C $(-1, 2)$

D $(2, -1)$

16 MCQ

Find the coordinates of the local minimum stationary point on the curve $y = 4x^3 - 48x + 9$ for $x > 0$ (Curve #8).



A $(-2, -55)$

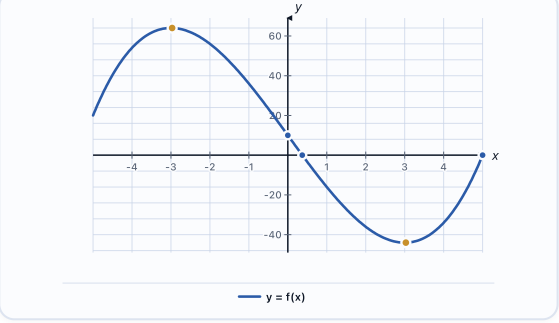
B $(4, -59)$

C $(2, 9)$

D $(2, -55)$

17 MCQ

Find the coordinates of the local minimum stationary point on the curve $y = x^3 - 27x + 10$ for $x > 0$ (Curve #9).



A $(3, -44)$

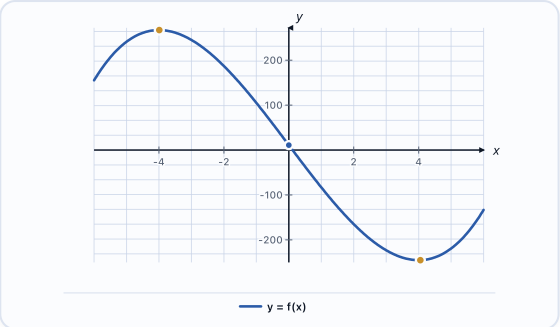
B $(3, 10)$

C $(6, -45)$

D $(-3, -44)$

18 MCQ

Find the coordinates of the local minimum stationary point on the curve $y = 2x^3 - 96x + 11$ for $x > 0$ (Curve #10).



A $(8, -247)$

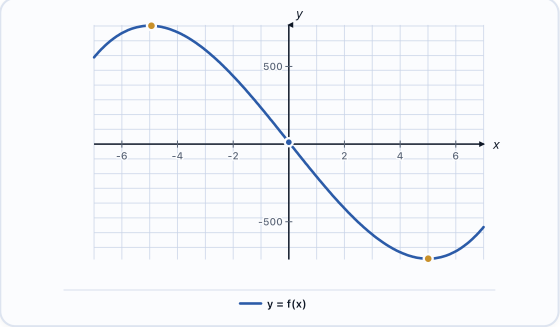
B $(4, -245)$

C $(4, 11)$

D $(-4, -245)$

19 MCQ

Find the coordinates of the local minimum stationary point on the curve $y = 3x^3 - 225x + 12$ for $x > 0$ (Curve #11).



A $(10, -741)$

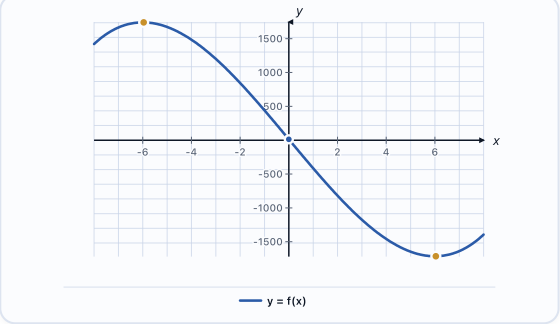
B $(5, -738)$

C $(5, 12)$

D $(-5, -738)$

20 MCQ

Find the coordinates of the local minimum stationary point on the curve $y = 4x^3 - 432x + 13$ for $x > 0$ (Curve #12).



A $(-6, -1715)$

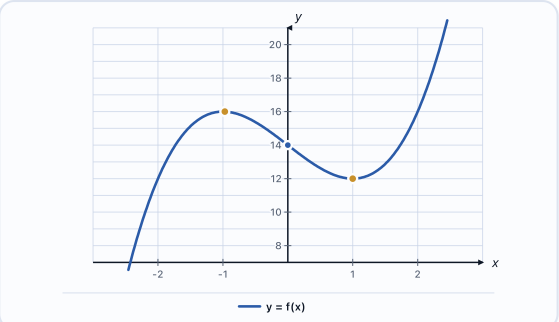
B $(12, -1719)$

C $(6, -1715)$

D $(6, 13)$

21 MCQ

Find the coordinates of the local minimum stationary point on the curve $y = x^3 - 3x + 14$ for $x > 0$ (Curve #13).



A $(1, 14)$

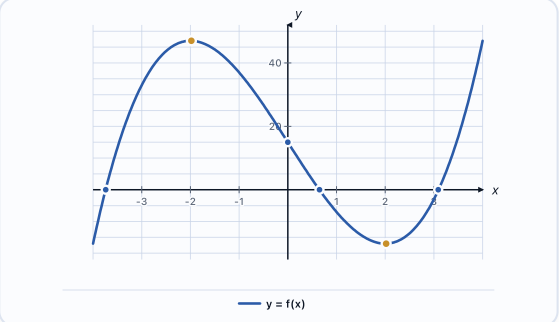
B $(-1, 12)$

C $(2, 11)$

D $(1, 12)$

22 MCQ

Find the coordinates of the local minimum stationary point on the curve $y = 2x^3 - 24x + 15$ for $x > 0$ (Curve #14).



A $(-2, -17)$

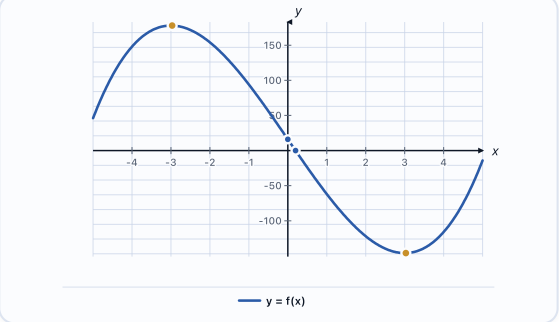
B $(4, -19)$

C $(2, -17)$

D $(2, 15)$

23 MCQ

Find the coordinates of the local minimum stationary point on the curve $y = 3x^3 - 81x + 16$ for $x > 0$ (Curve #15).



A $(3, -146)$

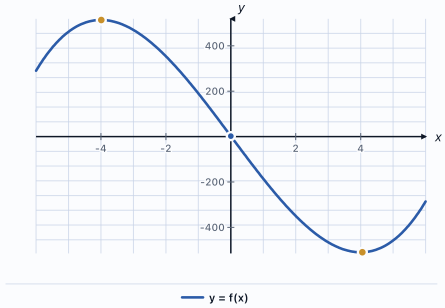
B $(-3, -146)$

C $(3, 16)$

D $(6, -149)$

24 MCQ

Find the coordinates of the local minimum stationary point on the curve $y = 4x^3 - 192x + 2$ for $x > 0$ (Curve #16).



A $(4, 2)$

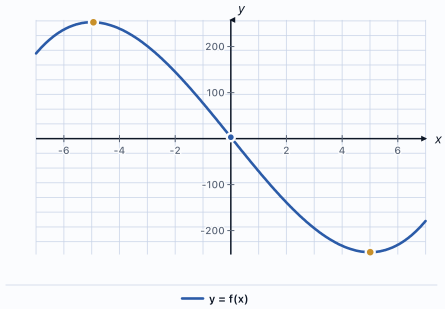
B $(4, -510)$

C $(-4, -510)$

D $(8, -514)$

25 MCQ

Find the coordinates of the local minimum stationary point on the curve $y = x^3 - 75x + 3$ for $x > 0$ (Curve #17).



A $(10, -248)$

B $(5, 3)$

C $(-5, -247)$

D $(5, -247)$